MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, RA, M650
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63142A (AR), dated 08 February, 1981, and is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: STANDARDS, MILITARY, add the following:

"MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification."

PAGE 2

2.1: Drawings, delete: "9288023 - Pellet Explosive"

3.2.2: line 1, delete: "grease or silicone" and substitute "lubricant".

Add the following new paragraph:

"3.2.4 Rotating band. The surface of the rotating band lands shall be examined for surface damage.

On the forward two lands, there shall be no evidence of raised copper due to nicks, digs or damage of any kind.

On the aft, three lands, any nicks or digs extending 1/2 the land width or less are acceptable, provided the maximum band diameter is not exceeded.

Surface scratches without evidence of raised copper are acceptable on all five lands."

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PAGE 5

3.3.2: Delete "pellet" and substitute "booster".

3.3.5: line 4, delete: "4.4.2.9" and substitute "4.5.4."

Add paragraphs 3.3.6 and 3.3.7 as follows:

"3.3.6 Metal parts defects. There shall be no crack, pipe, or thin wall section in the aft warhead.

3.3.7 Foreign material. There shall be no foreign material in the explosive charge."

PAGE 6

Add the following new paragraph:

"3.4.7 Interior metal defects. The aft warhead cavity shall be free from embedded foreign matter, raised material or sharp edges."

* 3.6.1.1 Add the following to end of paragraph:

"The rocket motor shall ignite not earlier than four seconds after firing. Lots with early motor ignitions, which pass all other proving ground test requirements, shall be referred to the technical agency for disposition."

PAGE 7

3.6.1.2.2: add the following: "unless the rocket motor fails to ignite."

Delete paragraph 3.6.2 in its entirety and substitute the following:

"3.6.2 Safety. There shall be no premature burst or indication of metal parts separation in the gun bore, or in flight, when fired at any of the conditions specified in 4.4.3.4. In addition, no projectile fired in the safety phase shall have a horizontal range less than 97.5 percent of the mean unless the rocket motor fails to ignite."

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3.10: line 2, delete: "in the exterior of the projectile" and substitute "in the nose threads or on the exterior of the aft warhead."

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4.3.1: Delete "Pellet, Explosive 9288023 10".

PAGE 10

Table II, delete: "Pellet, Explosive 4.4.2.7".

PAGE 11

4.4.1: Delete "h. Explosive.....manufacturer." and substitute:

"h. Booster assemblies PA-524 from one lot interfix from one manufacturer."

PAGE 12

4.4.2.1: cha ge title to read: "Delay Assy and Motor Body Prior to Assembly."

Change Drawing Number block to read: "9287970/9288025".

Major 103, delete in its entirety and substitute the following:

Major category:

"Major 103: Rocket motor body protective cap missing or improper 0.40% 3.2 Visual."

Add the following Major defect:

Major category:

"Major 104: Nozzle loose or missing 100% 3.2 Visual/Manual".

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4.4.2.3: Major 103, delete: "Grease or silicone" and substitute "Lubricant".

Major 104, delete: "Silicone compound" and substitute "Lubricant"

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4.4.2.4: Major 101, delete: "Test" and substitute "Visual".

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4.4.2.5: delete "Critical 1: Nozzle insert missing, loose, inverted, or damaged" and substitute the following:

"1. Nozzle insert missing, or damaged, after assembly of delay".

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Delete 4.4.2.7 in its entirety.

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4.4.2.8: Critical defect 2, delete Requirement Paragraph "3.4.6" and substitute "3.4.5".

Critical defect 3, delete Requirement Paragraph "3.4.8" and substitute "3.4.7".

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Major 102 and 103: Delete in their entirety and substitute the following:

Major category:

"Major 102: Forward protective cap missing or improper 0.40% 3.2 Visual.

Major 103: Aft protective cap missing or improper 0.40% 3.2 Visual".

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4.4.2.9: Critical defect 3, delete Requirement Paragraph " 3.4.7" and substitute "3.4.6".

Add the following critical defects:

"4. Presence of crack, pipe or thin wall section in the aft warhead 100% 3.3.6 4.4.3.1.2.

5. Foreign material in explosive charge 100% 3.3.7 4.4.3.1.3".

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Major category:

Major 102, Inspection Method column: Delete "Gage" and substitute "Visual".

Major 103: Delete "Explosive pellet" and substitute "Booster".

Major 104: Delete in its entirety.

Major 105, delete in its entirety.

Add the following Major defects:

Major category:

"Major 107: Booster cavity depth incorrect 0.40% 3.2 Gage.

Major 108: Booster cavity diameter incorrect 0.40% 3.2 Gage".

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4.4.2.10: delete: "Explosive pellet" and substitute "Booster".

PAGE 23

4.4.2.11: delete: "Disc" from title and substitute "Disk and Disc Assembly".

Add Major defects 102 and 103 as follows:

Major category:

"Major 102: Adhesive missing or insufficient on disk (disc) prior to assembly of disc (disk) 0.40% 3.2 Visual

Major 103: Disc not assembled concentric to disk (allowing improper seating in aft warhead) 100% 3.2 Visual/Templet".

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4.4.2.12: Major 102, delete: "Disc" and substitute "Disk, disc".

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4.4.2.13: Major defect 105, delete: "15" in the No. of Sample Units block.

Major defect 106, delete "15" in the No. of Sample Units block.

Add "0.40%" to the AQL block.

PAGE 26

4.4.2.14: Critical defect 1: Delete "O-ring groove".

Major 101: Delete in its entirety.

Major defect 106: Delete in its entirety and substitute the following:

Major category:

"Major 106: Forward grain cushion improper, missing or improperly lubricated, prior to application of sealant 0.40% 3.2 Visual"

Major defect 108: Delete in its entirety and substitute the following:

Major category:

"Major 108: Aft protective cap missing or improper on aft warhead 0.40% 3.2 Visual."

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4.4.2.15: Major 103, delete: "Manual/Visual" in the Inspection Method column and substitute "Gage".

Add "over maximum" after "rocket motor".

Major 105, delete "15" in the No. of Sample Units column.

Delete "4.3.3.2" in the Paragraph Reference/Inspection Method column and substitute "4.4.3.2".

Major 106, delete "3.2" in the Requirement paragraph and substitute "3.2.4".

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4.4.2.15: add the following two Minor defects:

Minor category:

"Minor 205: Lifting plug improperly lubricated 0.65% 3.2 Visual.

Minor 206: Warhead lot number not stamped on motor body rotating band 0.65% 3.2 Visual".

PAGE 29

4.4.2.16: Add Special Defect a. as follows:

a. Cover support missing (6 places) 100% 3.2 Visual"

Delete Major defect 103 and substitute the following:

"Major 103: Strapping or staples missing or loose 0.40% 3.2 Visual".

Minor Defect 203: Delete in its entirety and substitute the following:

"Minor 203: Strapping or staples improperly engaged 0.65% 3.2 Visual".

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Add 4.4.3.1.2 and 4.4.3.1.3 as follows:

"4.4.3.1.2 Metal parts defects. All radiographs of warhead section loading assemblies shall be examined for evidence of crack, pipe and thin wall sections in the aft warhead. The presence of any such indications shall cause that assembly to be classed defective and removed from the lot. (See Section 6.14).

4.4.3.1.3 Foreign material. All radiographs of warhead section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any foreign material shall cause that assembly to be classed defective and removed from the lot."

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4.4.3.2: delete in its entirety and substitute the following:

"4.4.3.2 Motor body - warhead assembly torque (see 3.5 - Major defect. The torque applied to the motor body-warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hours production for a check torque application. If any projectile fails to comply with the torque requirements, as evidenced by relative motion at the joint when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four-hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine display shall be checked during operation on a subsequent assembly. If the indicated torque is in or below the lower 20% of the acceptance range, a verification check shall be made. If the machine fails the verification, all projectiles assembled since the last acceptable verification shall be rejected."

PAGE 32

4.4.3.3.d: delete in its entirety and substitute the following:

"d. Voids or missing increments in the delay column."

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* 4.4.3.4: Delete in its entirety and substitute the following:

"4.4.3.4 Proving ground (see 3.6). These tests shall be performed as specified in 4.5.3. Defects shall be classified as follows:

Categories	Defects
Critical:	
1.	Premature burst or metal parts separation.
Special:	
a.	Range less than 97.5 percent of mean horizontal range unless rocket motor fails to ignite (see 6.16).
Major:	
101.	Range probable error exceeds allowable limit (see 3.6.1.2.1)
102.	Rocket motor fails to ignite.
Minor:	None defined".

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* 4.4.3.4.2.2: Delete "15 sample" and substitute "14 sample".

Add paragraph 4.4.3.4.2.3 as follows:

"4.4.3.4.2.3 Motor failure. Failure of a rocket motor to ignite during +145°F safety tests shall be considered a Major defect, and the lot shall be rejected. If a motor ignition failure occurs during -50°F safety tests, the lot shall be referred to the procuring agency for disposition (see 6.14)."

4.4.3.5: Delete in its entirety.

4.4.3.6: Delete in its entirety.

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4.4.3.7: delete in its entirety and substitute the following:

"4.4.3.7 Forward warhead - Aft warhead assembly torque (see 3.5). The torque applied to the forward warhead - aft warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hour's production for a check torque application. If any projectile fails to comply with the torque requirements, as evidenced by relative motion at the joint, when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four-hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine display shall be checked during operation on a subsequent assembly. If the indicated torque is in or below the lower 20% of the acceptance range, a verification check shall be made. If the machine fails the verification, all projectiles assembled since the last acceptable verification shall be rejected."

4.4.3.8: delete the following sentence: "Fifteen (15) projectiles shall be randomly selected for this test."

4.4.3.9: line 1, delete: "Eight projectiles" and substitute "Six aft warheads".

Line 3, delete: "Any projectile" and substitute "Any aft warhead".

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4.5.1.1: lines 25 and 26, delete: "unless identified as extraneous artifacts or metal parts irregularities" and substitute "unless identified as film or processing related or as acceptable metal parts irregularities."

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PAGE 36

4.5.1.2: delete the last sentence: "Beam alignment will be established to optimize detection of gap at aft end of propellant grain."

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4.5.1.6: first sentence, delete: "projectile" and substitute "aft warhead."

Last sentence, delete: "of each" which follows the words "In addition, one x-ray".

Add the following new paragraph:

"4.5.1.7 Operator qualifications. For operation of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For setup and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

4.5.2: line 5, delete: "rocket-off cap" and substitute "rocket cap".

PAGE 42

4.5.3.2: Change "-65°F" to read "-50°F", (two places).

4.5.4, lines 5 and 6, delete: "projectile" and substitute "aft warhead".

PAGE 43

4.5.5: delete "After touch-up...to maximum allowable diameter." and substitute:

"After touch-up and joint painting, projectiles shall be passing through a ring gage as specified on dwg. 9287994."

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4.5.6: delete in its entirety and substitute the following:

"4.5.6 Exudation. The exterior of the aft warhead shall be visually inspected for exudation. If no exudate is found, remove the forward protective plug and inspect for exudation visually."

* 6.3: delete: "ARRADCOM, Attn: DRDAR-QAR-I" and substitute "ARDEC, Attn: AMSMC-QAR-I (D)".

PAGE 44

* 6.7 Delete "*30 15" and "*60 25" and substitute "*30 14" and "*60 24".

PAGE 45

* 6.8: delete "ARRADCOM, DRDAR-QAR-Q, and DRDAR-LCU-CP" and substitute "ARDEC, AMSMC-QAR-Q (D), and SMCAR-LCU-SA".

* 6.9: delete: "ARRADCOM, DRDAR-QAR-Q and substitute "ARDEC and AMSMC-QAR-Q (D)".

6.11: Delete in its entirety.

* 6.12: delete: "ARRADCOM" at the end of paragraph and substitute "ARDEC".

Add paragraph 6.14 as follows:

"6.14 Rocket motor ignition failure at -50°F. Degradation of rocket motor ignition reliability is expected at temperatures below -25°F. When an ignition failure occurs at -50°F during acceptance testing, the procuring agency must evaluate the following prior to determining lot disposition:

a. The delay assembly shall have functioned properly as evidenced by review of radar data.

b. The failure shall not be related to a change in components or processes.

c. The effect on compatibility with the M753 NUC round must be considered. This may require a retest at -30°F.

d. The rate of failure should not be significantly different from prior experience."

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* Add the following new paragraphs:

"6.15 Notification of defects. The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of radiographs. The rejected projectile will be held pending disposition by the procuring activity.

6.16 Short range. Projectiles failing to achieve required range with proper motor function shall be considered critical defects. Projectiles failing to achieve required range due to improper motor ignition delay or motor burn shall be considered as major defects."

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